TP°4 : ARCHITECTURE ORIENTÉE SERVICES (SOA)

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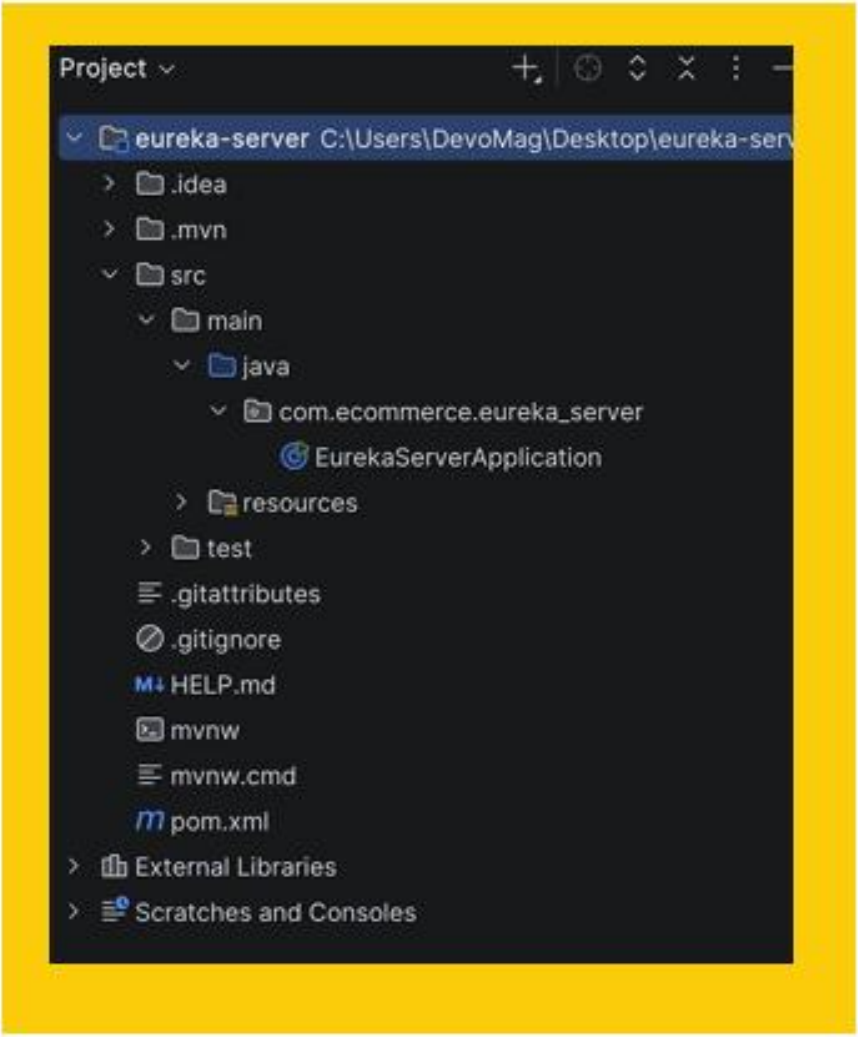
Encadré par : Pr. BE ELBAGHAZAOUI

Année Universitaire : 2025 – 2026

Étape 1: Création du Serveur Eureka (L'annuaire)

Étape 2: Configuration d' Eureka

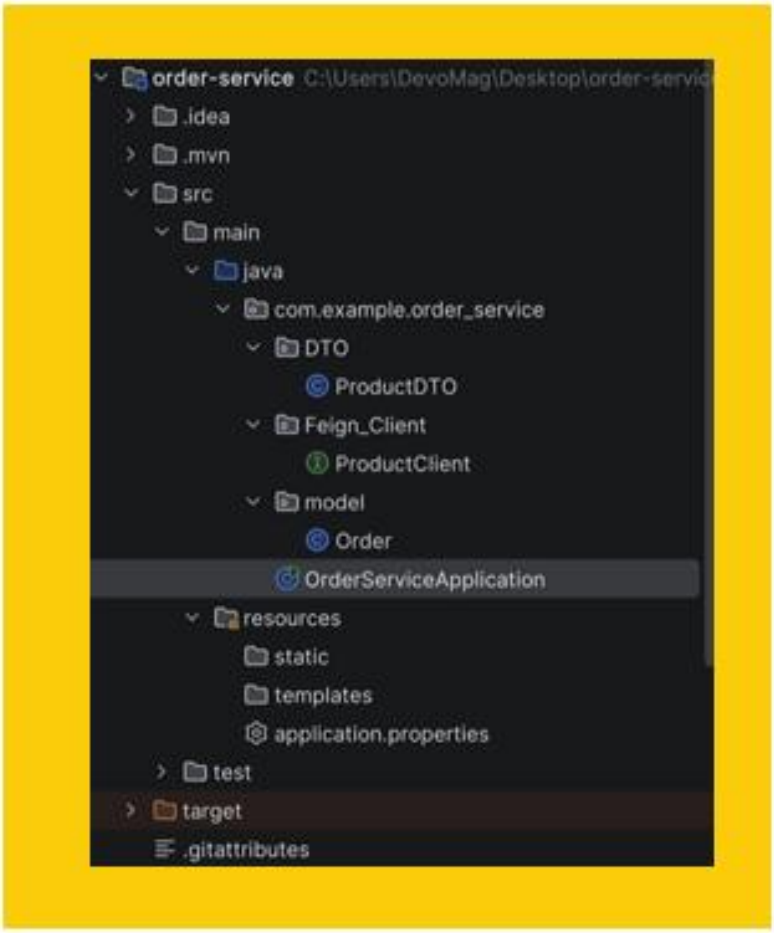


Démarrage

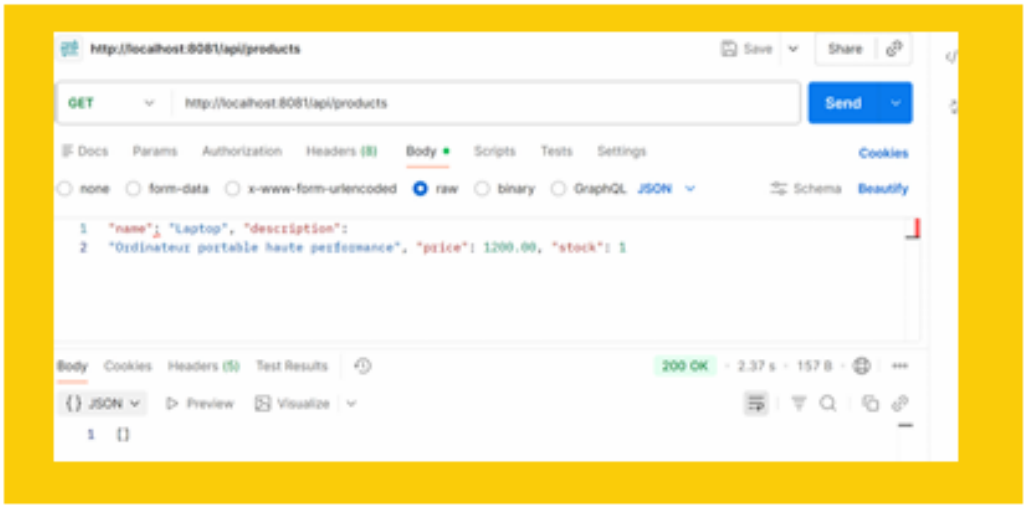
Étape 3: Créer le service Product

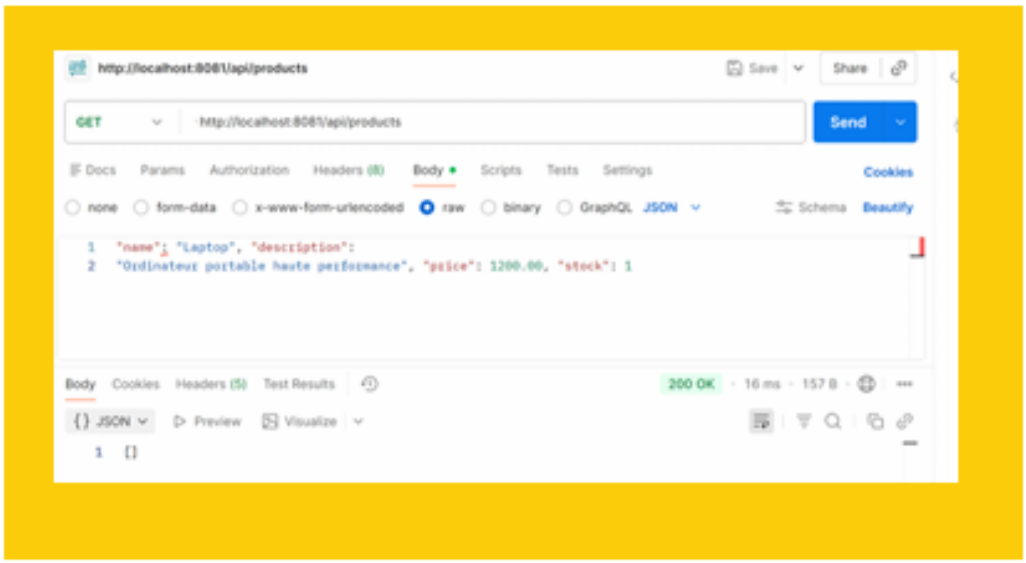
Étape 4 : Créer le service Order

Étape 5 : Tester l'application

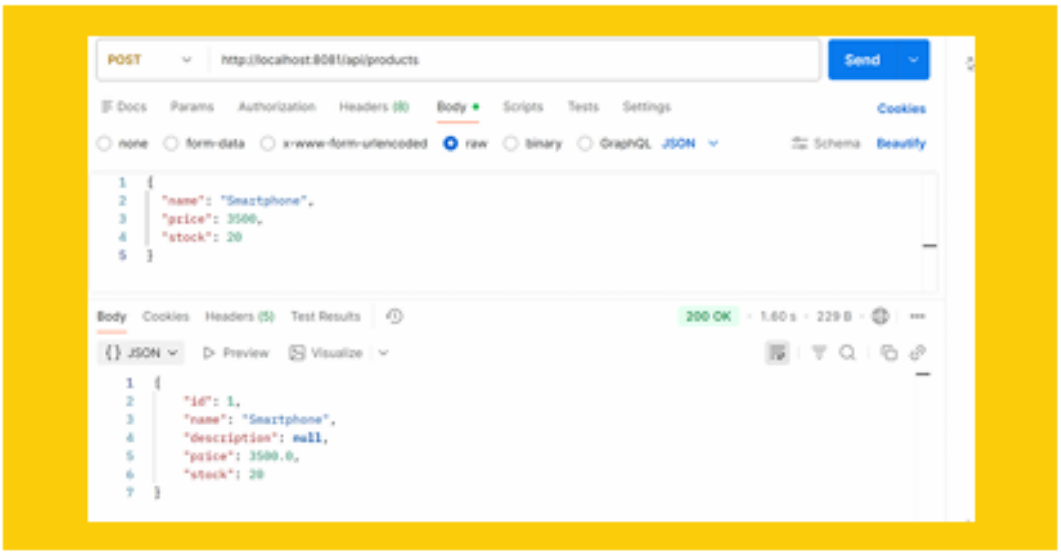


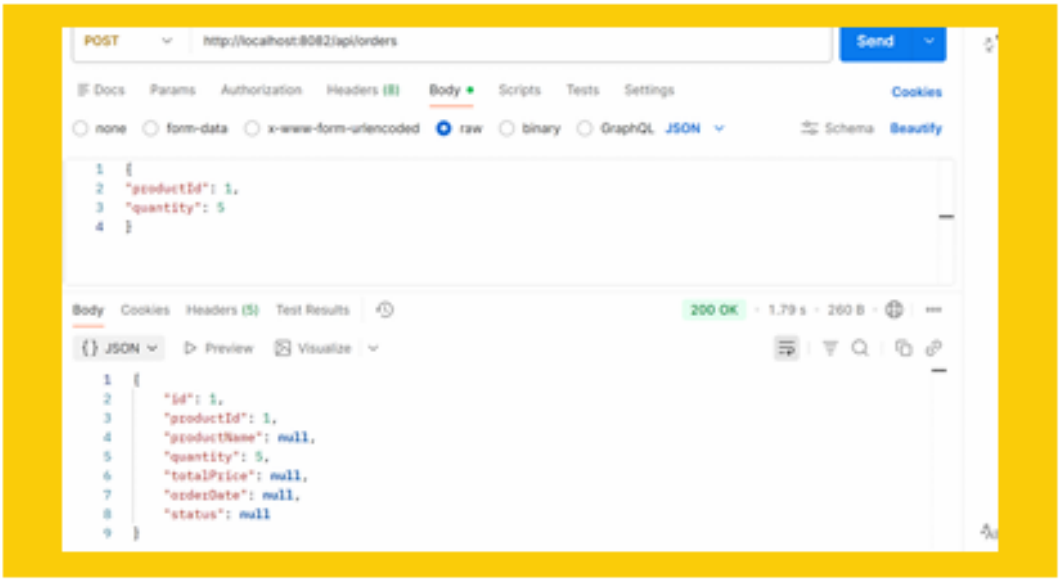
Tests avec Postman



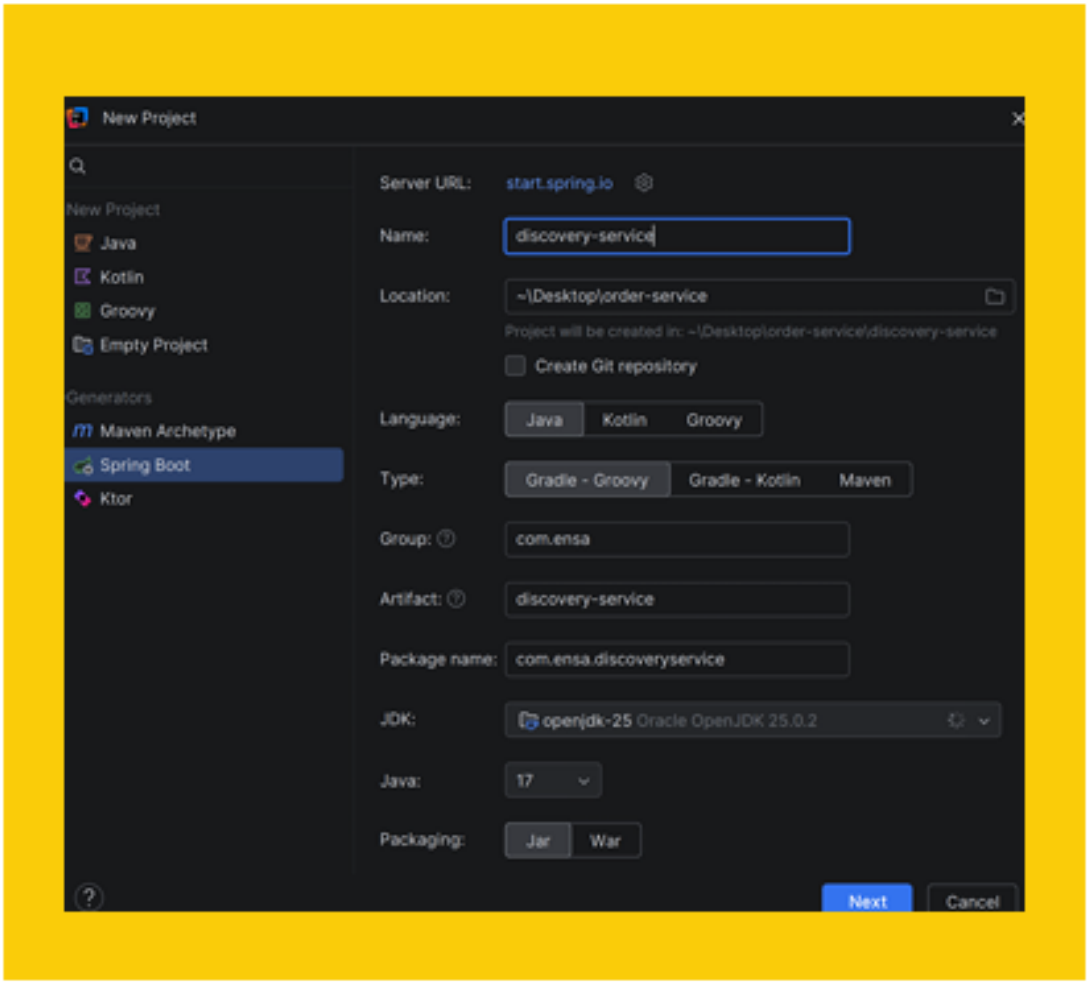


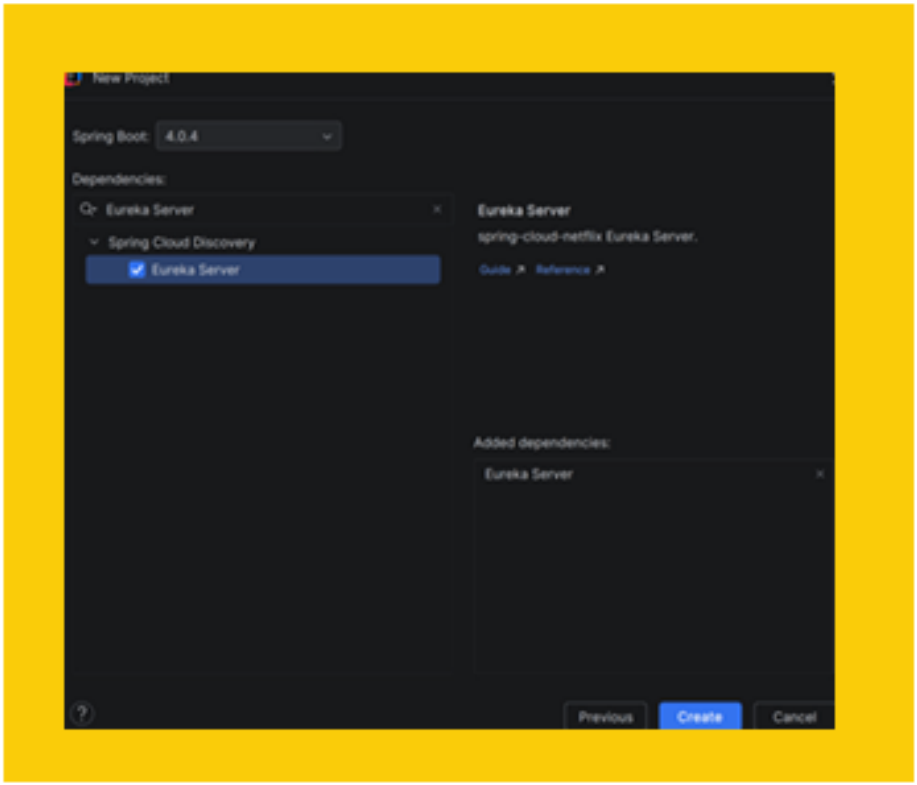
EXERCICE1

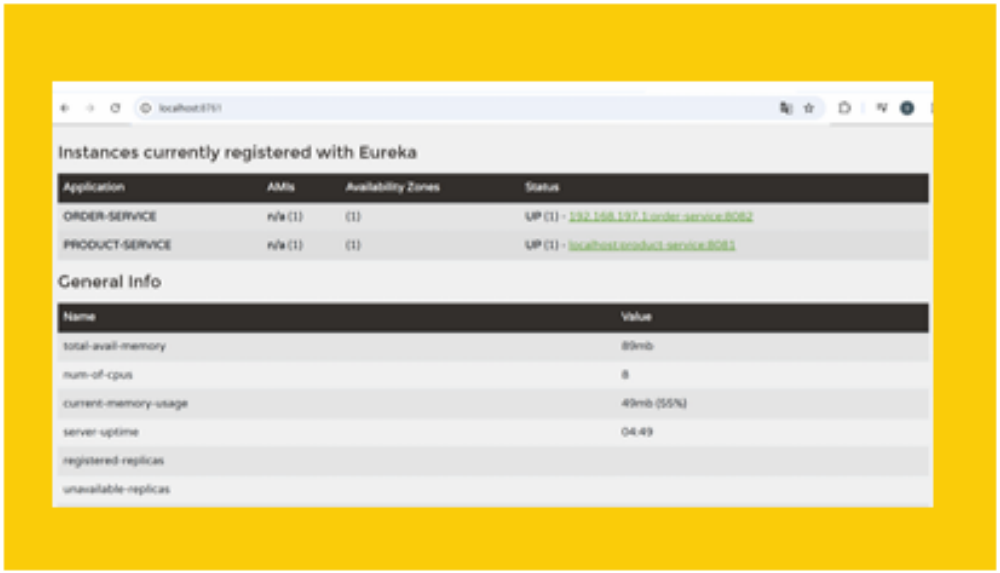


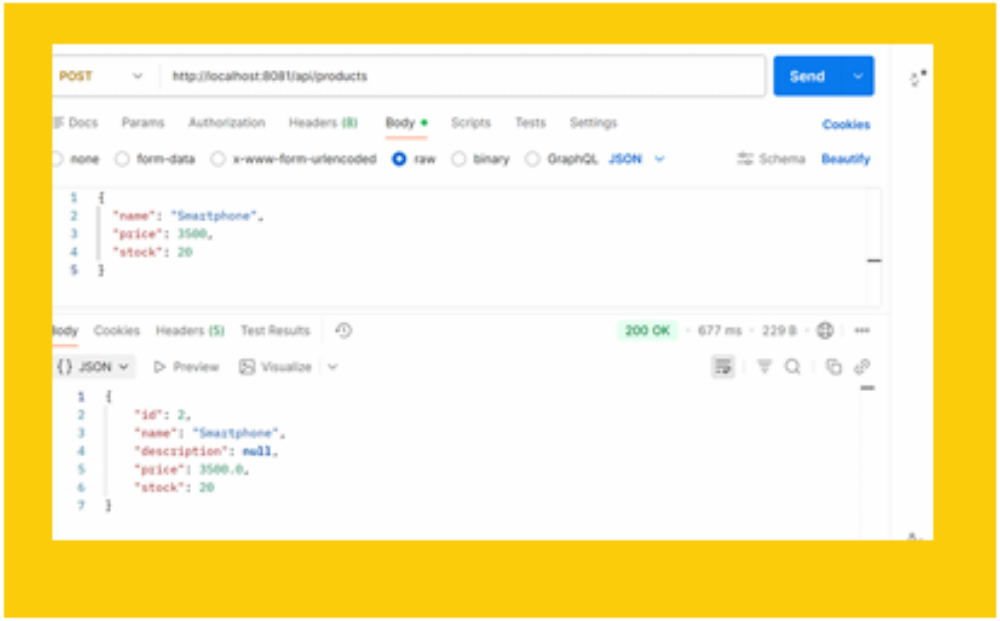


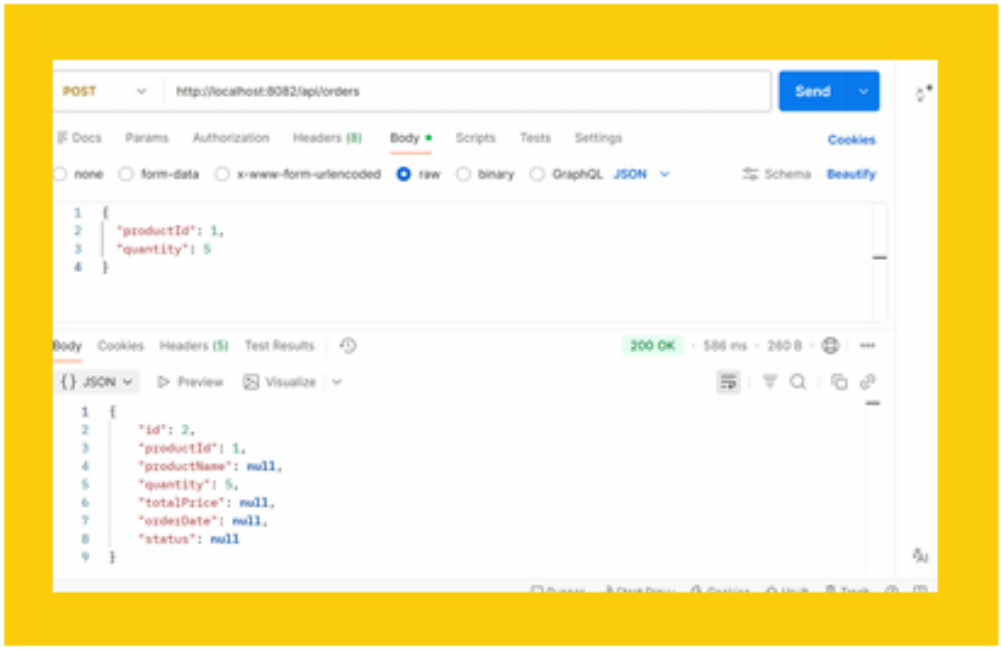
EXERCICE2

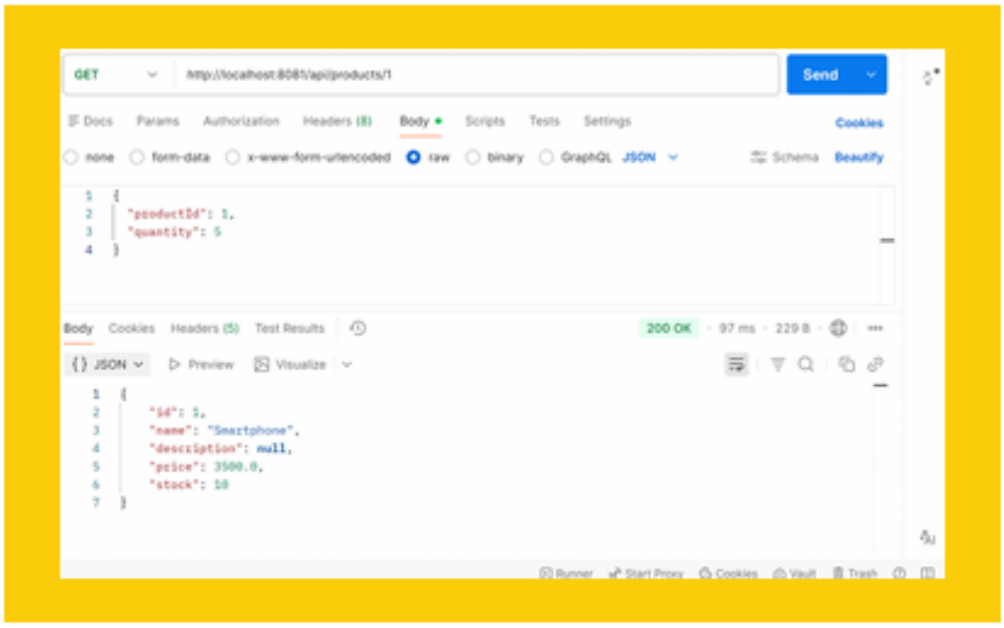

New Project
New Project
Java
Kotlin
Groovy
Empty Project
Generators
Maven Archetype
Spring Boot
Ktor
Server URL:
start.spring.io
Name:
discovery-service
Location:
~\Desktop\order-service
Project will be created in: ~\Desktop\order-service\discovery-service
Create Git repository
Language:
Java
Kotlin
Groovy
Type:
Gradle - Groovy
Gradle - Kotlin
Maven
Group:
com.ensa
Artifact:
discovery-service
Package name:
com.ensa.discoveryservice
JDK:
openjdk-25 Oracle OpenJDK 25.0.2
Java:
17
Packaging:
Jar
War
Next
Cancel


New Project
Spring Boot:
4.0.4
Dependencies:
Eureka Server
Spring Cloud Discovery
Eureka Server
Eureka Server
spring-cloud-netflix Eureka Server.
Guide
Reference
Added dependencies:
Eureka Server
Previous
Create
Cancel

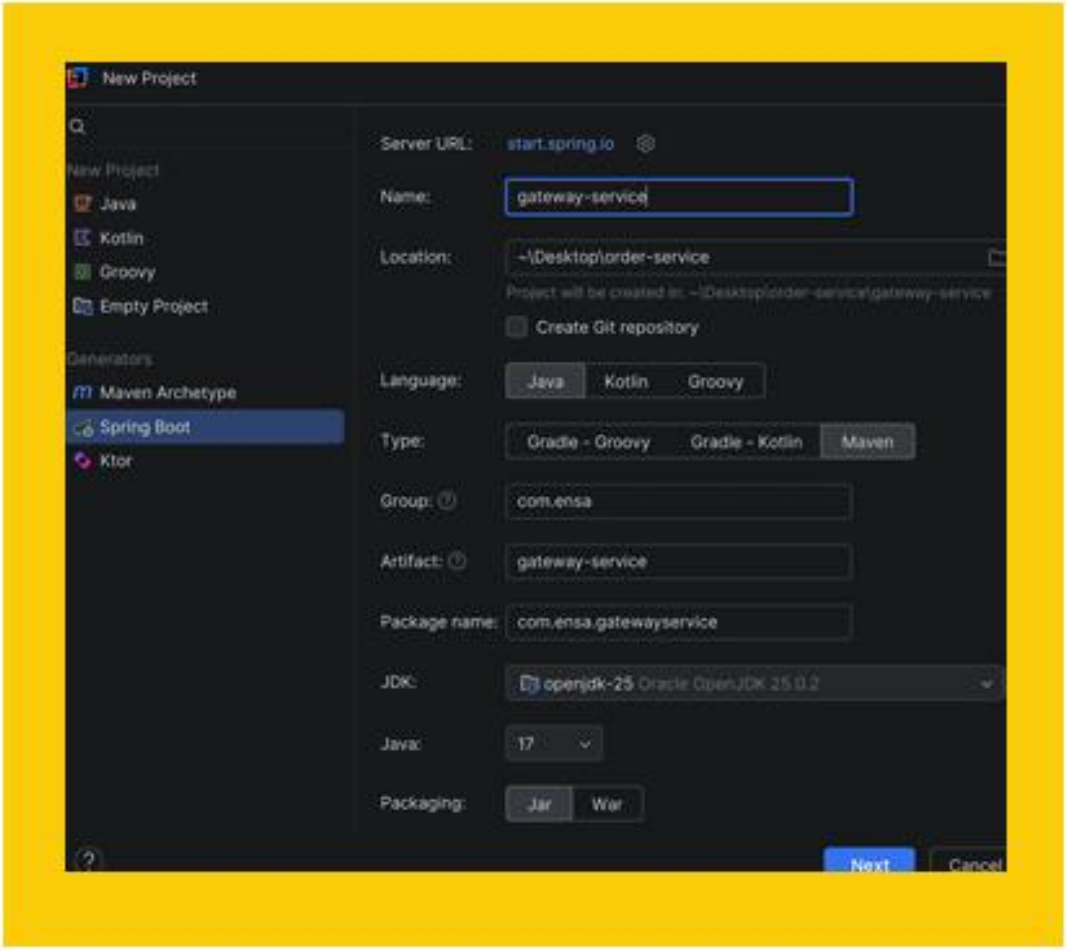

localhost:8761
Instances currently registered with Eureka
Application
AMIs
Availability Zones
Status
ORDER-SERVICE
n/a (1)
(1)
UP (1) - 192.168.197.1:order-service:8082
PRODUCT-SERVICE
n/a (1)
(1)
UP (1) - localhost:product-service:8081
General Info
Name
Value
total-avail-memory
89mb
num-of-cpus
8
current-memory-usage
49mb (55%)
server-uptime
04:49
registered-replicas
unavailable-replicas

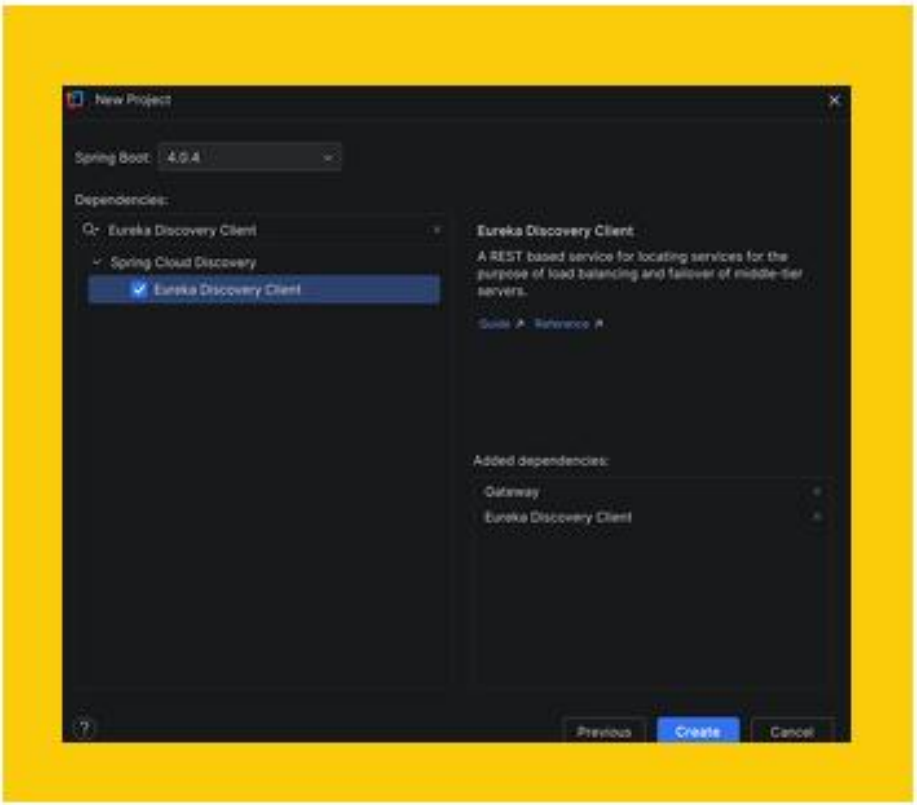

POST
http://localhost:8081/api/products
Send
Docs
Params
Authorization
Headers (8)
Body
Scripts
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"name": "Smartphone",
"price": 3500,
"stock": 20
}
Body
Cookies
Headers (5)
Test Results
200 OK
677 ms
229 B
JSON
Preview
Visualize
{
"id": 2,
"name": "Smartphone",
"description": null,
"price": 3500.0,
"stock": 20
}

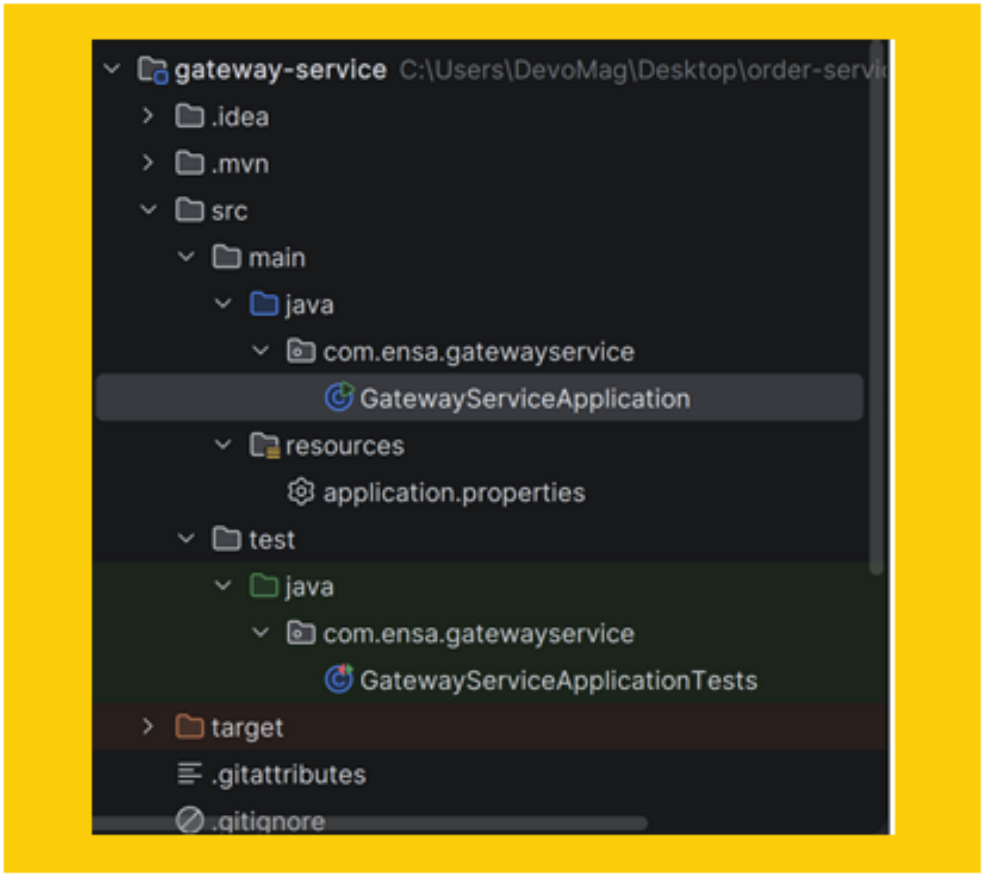

POST
http://localhost:8082/api/orders
Send
Docs
Params
Authorization
Headers (8)
Body
Scripts
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"productId": 1,
"quantity": 5
}
Body
Cookies
Headers (5)
Test Results
200 OK
586 ms
260 B
JSON
Preview
Visualize
{
"id": 2,
"productId": 1,
"productName": null,
"quantity": 5,
"totalPrice": null,
"orderDate": null,
"status": null
}

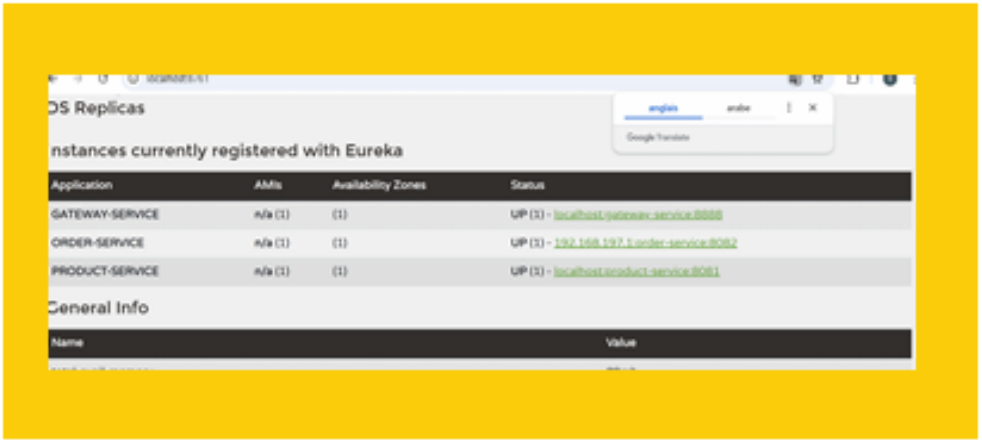

GET
http://localhost:8081/api/products/1
Send
Docs
Params
Authorization
Headers (8)
Body
Scripts
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"productId": 1,
"quantity": 5
}
Body
Cookies
Headers (5)
Test Results
200 OK
97 ms
229 B
JSON
Preview
Visualize
{
"id": 1,
"name": "Smartphone",
"description": null,
"price": 3500.0,
"stock": 10
}
Runner
Start Proxy
Cookies
Vault
Trash

EXERCIE 3


New Project
New Project
Java
Kotlin
Groovy
Empty Project
Generators
Maven Archetype
Spring Boot
Ktor
Server URL:
start.spring.io
Name:
gateway-service
Location:
~\Desktop\order-service
Project will be created in: ~\Desktop\order-service\gateway-service
Create Git repository
Language:
Java
Kotlin
Groovy
Type:
Gradle - Groovy
Gradle - Kotlin
Maven
Group:
com.ensa
Artifact:
gateway-service
Package name:
com.ensa.gatewayservice
JDK:
openjdk-25 Oracle OpenJDK 25.0.2
Java:
17
Packaging:
Jar
War
Next
Cancel


New Project
Spring Boot:
4.0.4
Dependencies:
Eureka Discovery Client
Spring Cloud Discovery
Eureka Discovery Client
Eureka Discovery Client
A REST based service for locating services for the purpose of load balancing and failover of middle-tier servers.
Guide
Reference
Added dependencies:
Gateway
Eureka Discovery Client
Previous
Create
Cancel





Ce projet a permis de mettre en place une architecture microservices complète en utilisant Spring Cloud. L'intégration d'Eureka pour la découverte de services, de Feign Client pour la communication inter-services, et d'une API Gateway pour le routage centralisé, garantit un système dynamique, scalable et facile à maintenir. Cette approche moderne assure une séparation efficace des responsabilités tout en offrant une grande flexibilité d'évolution.